

# Website Traffic Analysis

Project summary in plain language: what we looked at, what we found, and what to do next

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## 1. What this project is about

Every website gets visitors, but not all visitors are equally valuable. Some arrive, look around and stay. Others leave within seconds. This project studies the visit records of one website to answer simple questions: **Where do visitors come from? Which sources bring the most interested visitors? And at what time of day is the website busiest?**

The answers help a website owner decide where to spend time and money, such as which sources to grow, which to fix, and when to publish new content.

## 2. The data in simple words

The data is a record of website visits, saved hour by hour (April to May 2024). It was exported from a website analytics tool and needed cleaning before use. The header row was misplaced, the column names were unclear and the numbers were stored as text. After cleaning, no values were missing and nothing was duplicated.

**Key terms.** A **visit** (session) is one trip to the website. An **engaged visit** is one where the visitor interacted rather than leaving straight away. A **channel** is the route a visitor took to arrive, such as search, social media or a link on another site.

**3,182**

hourly records

**7**

traffic sources

**162,895**

total visits

**55.3%**

visits were engaged

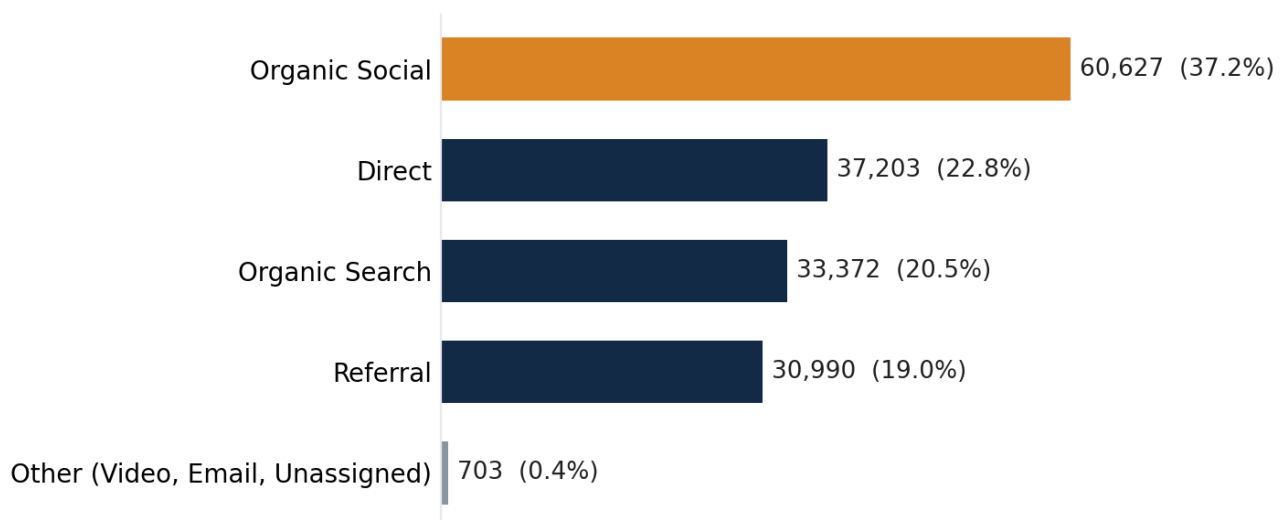
| Column in the data | What it means                                                                          |
|--------------------|----------------------------------------------------------------------------------------|
| Channel group      | Where the visitor came from (for example Google search, social media, another website) |
| Date and hour      | The day and hour the visits happened                                                   |
| Users              | How many different people visited in that hour                                         |
| Sessions           | How many visits happened in that hour (one person can visit more than once)            |
| Engaged sessions   | Visits where the person actually interacted with the site, not just opened it and left |
| Engagement rate    | The share of visits that were engaged, for example 60% means 6 out of every 10 visits  |

| Column in the data               | What it means                                                                  |
|----------------------------------|--------------------------------------------------------------------------------|
| Average engagement time          | How long a visitor stayed active on the site, in seconds                       |
| Events per session / Event count | How many actions (clicks, scrolls, page views) happened per visit and in total |

### 3. What we found

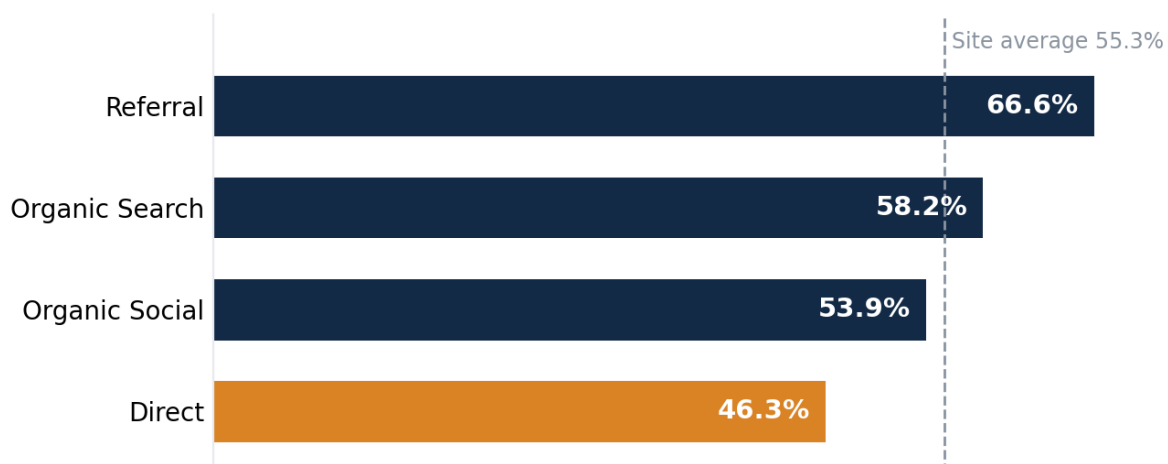
#### Finding 1: Social media brings the most visitors

More than a third of all visits (37.2%) came from **Organic Social**, meaning unpaid social media posts and shares. Direct visits (people typing the address or using a bookmark), search engines and links on other websites each bring roughly a fifth. Video, email and unlabelled sources together add up to less than 1%.



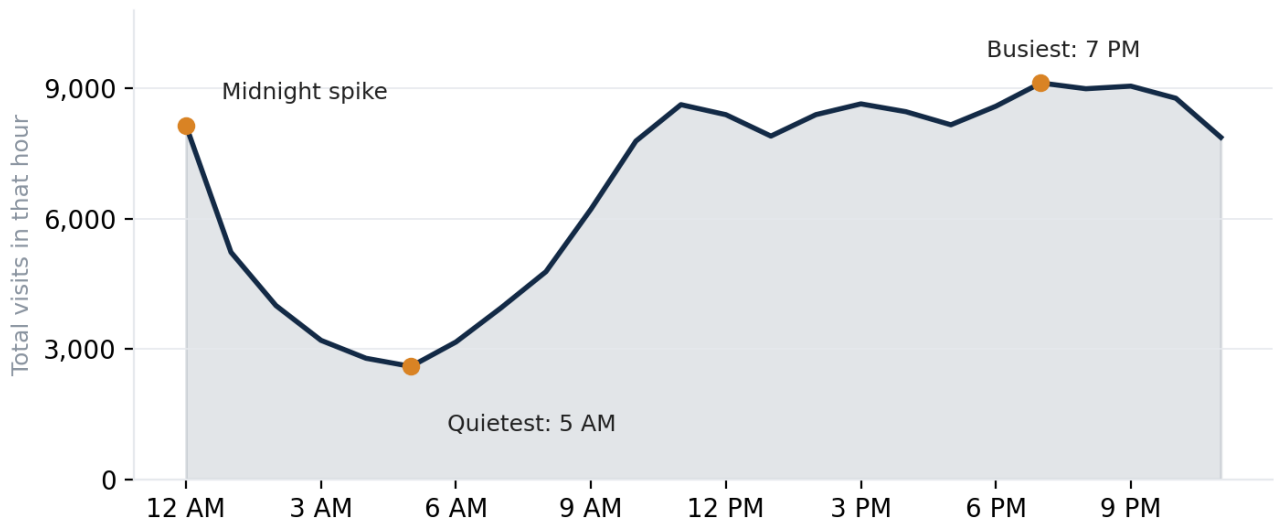
#### Finding 2: Visitors from other websites are the most interested

Having many visitors is not the same as having good visitors. **Referral** visitors, who arrive by clicking a link on another website, are the most engaged: **about 2 in 3 visits (66.6%)** were engaged, and they stayed about 90 seconds on average. Visitors from Direct, Search and Social stayed roughly 45 to 53 seconds. **Direct** is the weakest: fewer than half of its visits (46.3%) were engaged.



### Finding 3: The website is busiest in the evening and around midnight

Visits drop to their lowest at **5 AM (2,598 visits)**, climb quickly through the morning, stay high all afternoon and peak at **7 PM (9,129 visits)**. There is also a surprising jump at **midnight (8,148 visits)**, mostly from social media. Each source has its own busiest hour: Direct at 11 PM, Search at 2 PM, Social at midnight, and Referral stays steady from late morning to around 10 PM.



## 4. Things to keep in mind

- **Some sources are too small to judge.** Email had only 3 visits and Video only 141, so their numbers should not be trusted yet.
- **Unlabelled traffic looks like a tracking problem.** The 'Unassigned' group had almost no engaged visits (0.7%), which usually means the links are not tagged properly.
- **'Users' cannot simply be added up.** Because the data is hourly, the same person can be counted again in a later hour, so this report compares visits, which are safe to add.
- **One extreme value.** One record shows a visitor staying about 75 minutes (4,525 seconds), far above the typical 49 seconds, so averages involving time should be read as a rough guide.

## 5. What we suggest doing

1. **Grow links from other websites.** They bring the most interested visitors but only 19% of visits. Partnerships, guest articles and mentions are the most valuable way to grow.
2. **Improve the first impression for Direct visitors.** More than half leave without engaging. Faster pages, a clearer home page and obvious next steps can help.
3. **Make social media visitors stay.** Social is the biggest source, but its engagement (53.9%) is a little below the site average (55.3%). Make sure each post leads to a page that matches what it promised.
4. **Publish at the right time.** Schedule posts and announcements for the 7 to 9 PM window and again near midnight, when the most people are online.
5. **Fix the tracking gaps.** Tag all campaign links, especially emails, so every visit is credited to the right source.

6. **Collect more data on small sources.** Run email and video campaigns for a few weeks before deciding whether to invest further.

## 6. Conclusion

The website has two different strengths. **Social media brings the crowd**, and **other websites bring the most interested visitors**. The weakest source is Direct traffic, where more visits end quickly than engage. Traffic follows a clear daily rhythm, quiet in the early morning and busiest in the evening. The best next steps are to grow referral links, improve the Direct experience, post at peak hours and fix the tracking so no visits go unlabelled.

## 7. Behind the scenes

The analysis was done in Python using the Pandas, NumPy, Matplotlib and Seaborn libraries. The full notebook, data and code are available at [github.com/Satyajeet745/Website\\_Analysis\\_project](https://github.com/Satyajeet745/Website_Analysis_project).

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